01 · LOT TILE: Product Plan

Doc: LOT-HW-01-PLAN · **Rev:** A (2026-11-14) · **Author:** Vadik, Inventor, COSMO® CIA (with Claude Code)

1. WHAT IT IS

LOT TILE is a pocket or desk object. It is the physical end of LOT-systems.com.

- The site's AI (Memory Engine, Quantum Intent Engine, contextual prompts) decides when the user should get a nudge and writes it as one short line, for example *"Coffee time!"*, *"Walk. 12°C, clear."* or *"Evening check-in."*
- The tile vibrates once, like a pager, and shows the line.
- **Copy** button, short press: the line is written to the user's **Log** tab on lot-systems.com as a `device_copy` entry. The screen then confirms with "Copied."
- **Copy** button, long press: the camera takes one frame, and the same Log entry gets a photo.
- Every 15 minutes the BME688 sensor reports local temperature, humidity, pressure and air quality. The site uses this in place of, or alongside, the city weather it fetches today (`/api/weather`).

The design follows the LOT style guide: no decorative colour, no emoji, "Done." instead of a check mark, and one primary action.

2. REQUIREMENTS TRACEABILITY (THE 19 POINTS OF THE BRIEF)

#	Brief	Design decision	Doc
1	PCBWay	PCBWay for PCB fabrication, PCBA (turnkey parts), CNC stainless enclosure and mirror polish, from one vendor	07
2	Pager-like notification from the AI site	Server pushes a ≤ 48-character line. Device: LRA haptic pulse, then the line on the Memory LCD	04, 06
3	Two-part stainless body	Part A: back tray (floor and walls, 316L). Part B: front bezel plate (316L) holding the cover glass. Joined with 4 × M1.0 Torx screws and a gasket	03
4	Flat silver square 40 × 40 mm, 5 mm high	40.0 × 40.0 mm footprint. **5.0 mm is achievable only with a chip-down MCU and a custom 1.6 mm cell** (DVT). EVT build is 6.5 mm	03 §3
5	Camera	Ultra-thin FPC VGA module (GC032A / GC0308 class, ≤ 2.5 mm Z), DVP to ESP32-S3	04
6	LOT API connector	New `/api/device/*` namespace with device-token auth and pairing by short code	06
7	Result in PDF manuals	`tools/build_pdfs.py` renders every doc into `manuals/*.pdf`	manuals/
8	Compress the information in each session	(a) Wire format: compact JSON lines ≤ 512 B, with deltas only. (b) Server condenses AI context into one pager line. (c) Each work session ends with a compressed ledger block in the session report	06 §7, report
9	Firmware documents	Firmware architecture, state machine and OTA doc, plus source in `firmware/`	05
10	Software to connect with firmware	`software/lotctl.py`: USB-serial provisioning (Wi-Fi and pairing), log tail, and a device simulator	05 §8
11	Separate documents	One subject per file, 01–09, each also built as its own PDF	README
12	Charger	Qi 5 W charging pad (off-the-shelf in EVT, LOT-branded stainless puck in PVT). USB-C on the pad only	04 §4, 07

#	Brief	Design decision	Doc
13	100-unit run	BOM costed at 100 units. PCBWay PCBA plus CNC lot. FCC path for a pre-certified module	02, 07
14	Weather sensor	Bosch BME688 (T, RH, P, VOC gas) behind a sintered vent with an ePTFE membrane	04
15	AI-grade off-the-shelf sensors	BME688 (on-sensor gas AI via BME AI-Studio), BMI270 IMU (on-chip gesture/wrist-wake AI), LTR-303 ambient light	02, 04
16	Copy button signals the Log tab	`POST /api/device/copy` creates Log `event: 'device_copy'`, which must be added to the Log tab's `displayableEvents`	06 §5
17	One side polished stainless steel	Back: mirror-polished 316L (#8 finish), with the Qi window in NCVM mirror-silver glass so the side still reads as uniform silver	03 §4
18	Other side: camera, screen, button	Front: 1.08" Memory LCD (top), camera (bottom left), Copy key (bottom right)	03 §2
19	Wireless charger	Qi RX, TI BQ51050B (receiver and Li-ion charger in one chip), with a 20 × 20 mm receiver coil behind the back window	04 §4

3. THE ONE PHYSICS CONFLICT TO RESOLVE

Solid stainless steel blocks wireless charging. A Qi field induces eddy currents in a steel back, which heats the steel and stops charging. Points 17 and 19 conflict if the back is one solid steel plate.

Proposed resolution (default, **decision D-1**): cut a Ø 26 mm window in the back plate and fill it with a flush cover of chemically strengthened glass with **NCVM coating** (non-conductive vacuum metallization). This is the finish phones use to look metallic while staying RF and induction transparent. The back still reads as one mirror-silver face, and the window shows only as a thin seam ring.

Alternatives, if Vadik prefers an all-steel back:

Option	Look	Charging	Cost / risk
A. NCVM glass window (default)	Silver, faint seam ring	Qi 5 W (practical 1–2 W)	Low
B. All-steel back, magnetic pogo pins	Pure polished steel, 2 pads visible	Wired via dock, fast	Low, but it is not "wireless"
C. All-steel back, charge face-down through the front glass	Pure back	Coil must sit under the display, which needs custom flex coil and display stack	High
D. Titanium or ceramic back	Different material	Qi works through zirconia	Changes brand material

4. PRODUCT DECISIONS TAKEN IN THIS PLAN

1. **MCU: ESP32-S3.** Wi-Fi and BLE, camera DVP interface, PSRAM for frames, mature Arduino/IDF, low cost. EVT uses the pre-certified `ESP32-S3-MINI-1U/-1-N4R2` module (FCC modular grant, 2.4 mm Z). The 5.0 mm DVT moves to chip-down `ESP32-S3FH4R2` (QFN56 7 × 7 × 0.9 mm). That needs our own RF layout and a full FCC Part 15C grant.
2. **Screen: Sharp Memory LCD, 1.08", 160 × 68 (LS011B7DH03).** Reflective silver-grey look, readable in sun, ~µW static draw (holds the last notification with power off the MCU) and about 1 mm thick. It fits the LOT monochrome aesthetic.
3. **Pager haptic:** 8 mm Z-axis LRA and TI DRV2605L (licensed haptic library, one-line "pager" waveform).
4. **Connectivity:** standalone Wi-Fi with BLE provisioning. No phone app is required in v1. Pairing uses a 6-character code entered on lot-systems.com.

5. **Delivery model:** the device wakes on a timer (default every 5 min, adaptive) and fetches pending notifications. Opening a socket for true push is a v2 option (power cost below).

5. OUT OF SCOPE FOR V1

Microphone and speaker, GPS, cellular, colour display, water resistance above IP54, any on-device generative AI (all AI runs on lot-systems.com).

6. SUCCESS CRITERIA FOR THE 100-UNIT RUN

- ≥ 95 % first-pass yield at final functional test
- ≥ 3 days battery at the default 5-minute cadence (EVT cell) and ≥ 2 days (DVT 5 mm cell)
- Notification latency ≤ cadence + 10 s. Copy-to-Log round trip ≤ 3 s on good Wi-Fi
- Back face passes cosmetic spec: #8 mirror, no visible tool marks at 30 cm under D65 light